# Geographies of Wellbeing in Poland (Urban-Rural, Regions, and Cities)

Sunday 18th October, 2026 14:53

## 1 changelog [proposal after this]

### 1.1 oct17

met with piotr incorporating his comments

big/broad items

- TODO rewrite text based on his comments, make text uptight
- add ad the end WP/tasks after timeline
- added "Work plan: key milestones" sec

specific items from comments in pdf

- style–made it more uptight/academic
- not convincing–added references to sen and seligman
- An US view. Simplistic.–ha! yes we are simplistic here :) i cut back
- Can be interesting but needs to be engaged in the current literature. Why this gradient is relevant in EU, Poland, WOrld - needs justification.–yes! done added soc lit; and elaborated on the gradient; and added diagram
- It is very descriptive. WHat is the hypohteiss here: is Poland follow a wider pattern with other Central Europe countries (perhaps a global pattern?)–no just that growth or shrinkage−− >happiness–explained better and more explicitly
- Good, but very descriptive. No breakthrough. A strong need for an innovative conceptual framework.–DONE added to hypotheses sec
- Looks as territorilly embedded. Dat harmonization seems problematic. It needs to be expained.–added re harmonization
- details needed.–added new sec "Work Plan: Key Milestones" and refered the reader to there
- Perhaps - discussion with the Catch-up lietarture of the Central EUrope? SOmehow it is a default underlying concept. POland compared to Other trasnsition coutries: Latin America? (See `https://books.google.com.fj/books?id=OBvQVhkIvdIC&printsec=frontcover`–added new hypothesis to compare with other eur countries; and added new hyopothesis on social groups: age and education–the educated and younger ones may fare better in cities v rural; and added a little bit on latin america/colombia too
- Perhaps engagement with the energy transformation in Europe which requires a massive change of consumption, would be important here as the large European policy altering lifestyles and consequently well-being.–yes!! added and citing my own paper on this!
- It requires explanation why citing so old fashioned/obsolete literature.–ok dropped Fromm

## 2 Background, Problem Description, and Objectives

The proposed project fits under the UN Sustainable Development Goals (SDGs): #3 "Ensure healthy lives and promote well-being for all at all ages," and #11 "Make cities and human settlements inclusive, safe, resilient and sustainable."

The proposed study's outcome of interest is multidimensional human wellbeing or flourishing. Two key dimensions are happiness or life satisfaction[1] and freedom (agency/autonomy).[2] Both measures are defined and described in data section.

Happiness is now recognized as a development goal in addition to GDP–the UN has passed resolution A/RES/65/309: "Happiness: towards a holistic approach to development" stating that "the pursuit of happiness is a fundamental human goal" and "recognizing that the gross domestic product indicator [GDP] by nature was not designed to and does not adequately reflect the happiness and well-being of people."[3]

Freedom (agency/autonomy) is a human right as stated by the Universal Declaration of Human Rights (art. 26, 29): "freedom for the realization of one's human potential." Freedom is an end-in-itself, as recognized by Amartya Sen (2000).

There is a new geography of wellbeing field (e.g., Ballas and Dorling 2013). The field is interdisciplinary with key contributions from sociology such as Ruut Veenhoven (Brulé and Veenhoven 2015, Veenhoven 1994, 2000b), Jan Delhey (Delhey and Newton 2005, 2003, Delhey and Dragolov 2016, Delhey 2004, Delhey et al. 2002, Alber et al. 2005) and David Bartram (Bartram 2013a, 2011, Hendriks and Bartram 2019, Bartram 2013b). The field has mostly international focus across countries, with only few recent studies focusing on specific countries (see next section).

While earlier there was a need for focus on economic growth in Poland, now there is a need for focus on wellbeing. In 2026 Poland has emerged as a one trillion dollar economy[4]. Polish PCGDP PPP[5] is very close to that of Japan, and Polish minimum wage of 31 PLN or 8.15 USD (as of 2026) is higher than the US minimum wage at 7.25 USD. Wellbeing is increasingly relevant over economic/survival basics such as income and unemployment in the past. When Poland was a developing country in the 90s, the focus rightly so was on survival and economic growth. As Poland has experienced economic miracle over past several years,[6] now it is time to focus on wellbeing. For visualization of the changing values from survival to wellbeing along with the economic growth see figure 1.

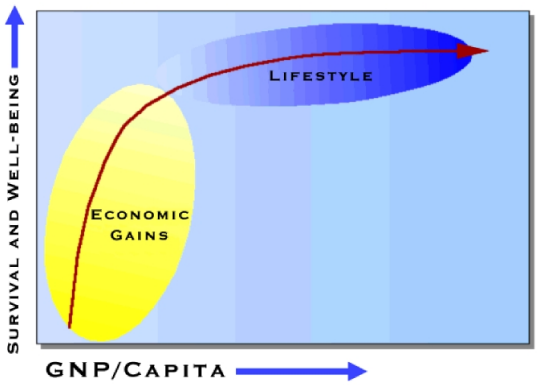



**Figure 1:** From survival to wellbeing as economic growth increases (Inglehart 1997). As Polish GNP/capita has increased sharply over the recent years, the scholarly (and public) focus can and should shift from survival to wellbeing.

Leading scholars advocate such change in focus from survival to flourishing. For instance, Seligman (Seligman 2004, Diener and

[1] We use the terms interchangeably as is the usual practice in the social indicators literature, but specifically we mean global cognitive evaluation of life as whole rather than momentary and fleeing affective happiness.

[2] Specifically we mean so called `freedom to`, not `freedom from` (Okulicz-Kozaryn 2014a). Poles are `free from`: live in a free country, have no coercion, and are free from restrictions and impediments. But are they `free to`? Are Poles actually able to choose, control and direct one's own life (Okulicz-Kozaryn 2015)?

[3] See `https://docs.un.org/en/A/RES/65/309`. Also see report promoting wellbeing as a human development measure by Nobel prize laureates Sen and Stiglitz (Stiglitz et al. 2009).

[4] `https://www.gov.pl/web/primeminister/poland-joins-the-trillionaires-club-a-historic-entry-into-the-worlds-top-20-economies`

[5] Per Capita Gross Domestic Product Purchasing Power Parity adjusted.

[6] For instance Warszawski Index Gieldowy (WIG) has increased about 120% over past 5 years.

Seligman 2004) led change in psychology that traditionally used to focus on the negative side of human condition such as loneliness, anxiety, and depression–where psychologists have been trying to bring people from failure to normal. He pointed out that we should now focus on how to go from normal to flourishing. Amartya Sen is another example (Stiglitz et al. 2009, Sen 2000). Being an economist, he pointed out the limits of economic measures such as income and production, and proposed that the time has come to focus on wellbeing. Poland still lacks a dedicated wellbeing study across its geographies and social groups. The proposed study aims to fill this gap.

## 3 State of The Art: Urban-Rural, Regions, and Cities

Poland is a diverse country in terms of its urban-rural, regional, and city geographies. Urban-rural is not a binary, but a gradient–we will use several steps to capture this gradient. Most scholars just use a binary, but wellbeing raises nonlinearly from urban to rural intesyfying at population size of below several hunderd thousand (Okulicz-Kozaryn 2017).

Urban-rural gradient matters a great deal, more than most scholars realize. Indeed, it is difficult to overestimate its importance–the key point is that urbanicity is not only a built environment, but a way of life (Wirth 1938). Classic sociologists were aware of the profound impact or urbanism, but modern empirical studies in this area do not properly ground their work in theory. The academic foundation of urban-rural wellbeing is sociological. Not only as urbanism as a way of life (Wirth 1938), but also based on the greatest classic writings in sociology: metropolis and mental life (Simmel 1903), and Gemeinschaft and Gesellschaft (Tönnies [1887] 2002). While there is a more recent empirical tradition in American urban sociology in study of urban wellbeing notably by Claude Fischer, this line of research has largerly been discontinued after 2000 (Fischer 1999, 1995, 1982, 1981, Fischer and Merton 1976, Fischer 1975, 1973, 1972). The proposed study aims to use the classic sociological theory with recent data to conduct a comprehensive study of the urban-rural happiness gradient in Poland.

Weckroth and Kemppainen (2021) is a closely related study to our proposed study–they analyze SWB by urban-rural using ESS in Europe–we will also use ESS among other datasets, but focus on Poland and add more recent waves and more wellbeing indicators. Knies (2026) is another close ESS study[7] –it uses multiple indicators, but only wave 6 and like other research does not focus on Poland. There are no studies on urban-rural freedom gradient, not just in Poland but anywhere–ours is the first.[8] Next we turn to regions.

Regions are an useful level of analysis as there are great differences within countries and across regions, and Poland is no exception. There are life satisfaction studies across regions, but no study focuses on Poland (e.g., Okulicz-Kozaryn 2011, Ballas and Rijnks 2023, Ziogas et al. 2023, 2020). Similarly, quality of life and wellbeing studies across regions and cities focus on cross-country patterns, not Poland, and tend to use older datasets (Okulicz-Kozaryn and Valente 2019, Okulicz-Kozaryn 2013). Poland has 16 voivodeships, so called NUTS2 level regions.[9] At NUTS2 level we will investigate geographic clustering and the relationship between objective development measures such as income and subjective measures (SWB). There are also interesting population dynamics at NUTS2 level, for instance, Mazovia (where Warsaw is located) has been growing, but most other regions have been stable or shrinking. We will investigate the effect of population change on wellbeing–we know that population dynamics are highly correlated with SWB (Okulicz-Kozaryn et al. 2024), but again, we do not know the relationship in Poland. The hypothesis is that population growth predicts greater wellbeing, and population shrinkage predict lower wellbeing. Population growth signifies bright outlook, and decline signifies difficult times coming. The theory that may explain the mechanism between population change and SWB is the so-called "tunnel effect." Humans tend to think that whatever is happening around, whether things are getting better or worse, will eventually happen to them

[7]And there are 2 more ESS studies: Ivanovic et al. (2022) is an urban-rural study, but again only 2 waves and no focus on Poland. Vaznonienė and Jarašiūnaitė-Fedosejeva (2019) is interesting ESS study in a sense that it does focus on one country and discusses country specific intricacies like us, but it is Lithuania. Okulicz-Kozaryn and Valente (2021) is a major recent study using WVS, but as most research it focuses on cross-country patterns, not Poland.

[8]Yamagishi et al. (2012) studies urban-rural freedom but in very different way from us–they show that urbanites are more likely to chose extravagant pens than rural folks. We will study freedom for urban-rural and regions (we do not have freedom variable for cities).

[9]Voivodeships=NUTS2, except Mazowieckie (the Warsaw region)–1 single voivodeship, but 2 NUTS2: Warsaw Metropolitan and Mazovian Region; so total of 17 NUTS2 regions.

as well.[10]

Finally, we have data representative of Bialystok, Gdansk, Cracow, and Warsaw with domain satisfactions (life, place, personal, and financial), and various quality of life indicators regarding amenities (e.g., public goods and services such as parks) and disamenities (e.g., community problems such as crime). We will investigate these across the cities. Specifically, large Warsaw (and perhaps Cracow) is likely to differ from small Bialystok. Warsaw stands out as not only the largest city, but also possibly a primate city, disproportionately larger and more influential than any other city.[11]

Last but not least, there are several related Polish studies in smaller local journals that are worth mentioning as they are mostly about Poland specifically–again, the international literature does not focus on Poland. We start with a hidden gem: Cichocki and Jabkowski (2009)–this is exactly the kind of study we want to conduct–just using more and newer data, controling for relevant predictors of SWB, and situating the study in the international literature. Sompolska-Rzechuła and Kurdyś-Kujawska (2020) is quite close, but uses different data: Polish Social Diagnosis dataset and does not control for SWB predictors. Vaznonienė and Wojewódzka-Wiewiórska (2021) uses ESS and studies urban-rural SWB like us, but focuses on Lithuania v Poland, and uses waves 4 and 9 only and 3 urbanicity categories only, and does not control for SWB predictors. Wojewódzka-Wiewiórska and Dudek (2023) is the same, just uses waves 2 and 8. Tobiasz-Adamczyk and Zawisza (2017) does the analyses across the regions and urban-rural, but their sample is elderly and the study is about health only. Stangierska et al. (2022) studies quality of life across green v not green urban areas, but only across Warsaw neighborhoods, and the sample size is small at 381 persons. Sompolska-Rzechuła and Kurdyś-Kujawska (2021) uses Polish Social Diagnosis and studies urban-rural SWB–but separately: predictors of SWB in urban separately from predictors of SWB in rural. Overall, while there are studies focusing on Poland, like our proposed study, they have important limitations as summarized above.[12] Hence, we know little about wellbeing geographies in Poland–we turn to our study.

# 4 Innovative Conceptual Framework And Hypotheses–Expected Progress Beyond the State the Art

It will be the first such study in Poland. The classic scholars of 19th and 20th century put forward research theories and theses that could not be properly tested at the time due to lack of data and computing. We ground our research better in classic theory, which allows us to pose research questions overlooked by the contemporary research. The innovative conceptal framework includes elaborated urban-rural gradient measurement. We will use 5-8 bins depending on the survey to measure urban-rural gradient, a critical improvement over usual binary or ternary urban-rural measurement in the literature. Urban-rural wellebing gradient is visualized in figure 2.

[10]

> Suppose that I drive through a two-lane tunnel, both lanes going in the same direction, and run into a serious traffic jam. No car moves in either lane as far as I can see (which is not very far). I am in the left lane and feel dejected. After a while the cars in the right lane begin to move. Naturally my spirits lift considerably, for I know the jam has been broken and that my lane's turn to move will surely come at any moment now. Even though I still sit still, I feel much better off than before because of the expectation that I shall soon be on the move (Hirschman, quoted in Ravallion and Lokshin 2000, p. 88).

[11] At $> 3m$ Warsaw metro is larger than Katowice metro at about 2.5m and third Karkow metro at 1.5m. While Katowice metro is not half or less of Warsaw metro as "the law of the primate city" would indicate (see discussion in Berry and Okulicz-Kozaryn 2012), it is only due to the location of Katowice in regional industrial cluster (Gornoslaski Okreg Przemyslowy (GOP)), Katowice city itself is much smaller than Warsaw at 0.28m v 1.9m. Maybe Warsaw is not such clear primate as Cairo (Mikhaeil et al. 2024), but it could be similar in its primate status to Bogota (Okulicz-Kozaryn et al. 2026).

[12] And there are several studies that differ from the above. Churski and Perdał (2022) has very fine geographical resolution of 380 units across Poland, but they do not study SWB, only the related objective indicators such as jobs. Weziak-Bialowolska et al. (2025) is one of the very many of the studies co-authored with VanderWeele, this time in Poland, but there is no geography in the study. Wkeziak-Bialowolska (2016) uses the same flash barometer 366 that we will use, but does not focus on Poland and instead on all European cities.

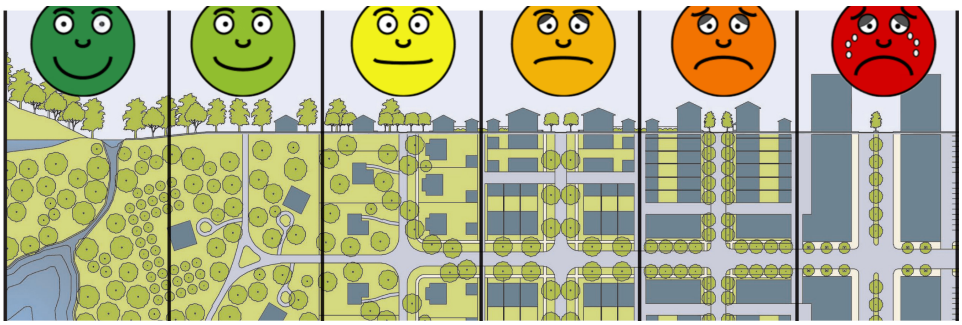


**Figure 2:** Urban-rural gradient visualization from right to left with transect diagrams sumbolizing urban-rural environments and corresponding frownie to smiley emojis symbolizing wellbeing gradient. From urban to rural there is increasing nature, biodiversity and hypothetiszed wellbeing. Source: Authors. The transect diagram for urbanicity is by Duany Plater-Zyberk & Company `https://transect.org/rural_img.html`

Another innovation is a more extensive measurement of wellbeing. We will use a battery of survey items to capture multidimensional wellbeing capturing not only life satisfaction as most literature does, but also functionings, agency, and vitality. Combining multistep urban-rural wellbeing gradient with multidimensional measurement of wellbeing and focus on Poland, our study is the first of its kind–nobody to date has used such a research design. Finally, our innovative conceptal framework includes using large multiple datasets at multiple levels: urban-rural (and over time since the 90s), voivodeship (NUTS2), and city (see next data section). The proposed research will test the following hypotheses to produce new knowledge. The new knowledge is not only desirable on its own (academic value), but it also has policy and practical implications–see section "Significance, Policy, and Practice."

**Q ??:** too many H? cut some? which ones? on the other hand this is for 4-5 papers over 4yrs so maybe ok?

$H_1$ Cities have lower multidimensional Subjective Wellbeing (SWB) than rural areas, with the largest drop occurring for the largest places. [WVS, ESS, FB539][13]

$H_1a$ Certain social groups such as working-age adults and higher social classes have higher SWB in cities v rural areas. [WVS, ESS]

$H_1b$ Over time, from 1989 to 2012, the SWB gap between cities and rural places has increased. [WVS]

$H_1c$ In a comparative perspective to other European countries, the urban-rural SWB gradient is steeper in richer and faster growing economically countries. [EVS, WVS]

$H_2$ More remote, less populated, and lower density regions have higher SWB than more central, more populated and higher density regions. [ESS, FB539]

$H_2a$ Regions that experienced population growth will have higher SWB than flat or shrinking regions. [ESS, FB539]

$H_2b$ Across regions correlation between objective quality of life measures such as income with SWB measures will be moderate only at $< 0.5$ thus revealing inadequacy of traditional progress/development measures such as income. [ESS, FB539]

$H_2c$ Urban-rural SWB gradient across regions will be relatively stable, i.e., larger places have higher SWB than smaller places. [ESS, FB539]

$H_3$ The relationship between quality of life (local amenities and disamenities) and domain satisfactions (life, place, personal, and financial) will differ across Bialystok, Gdansk, Cracow, and Warsaw. [FB366,419]

## 5 Data, Measurement, and Methods

The goal is to do a comprehensive study of Polish geographies of wellbeing, hence we use multiple datasets, each having its own strengths: World Values Survey (WVS), European Social Survey (ESS), and Flash Barometers (FB) for regions (FB539) and cities

[13]WVS=World Values Survey; ESS=European Social Survey; FB539=Flash Barometer #539–these and other datasets are described in the next section.

(FB366,419).

WVS has a usual SWB item: "All things considered, how satisfied are you with your life as a whole these days?" on scale 1-10. Such SWB measures tend to be reasonably valid and reliable (Diener 2009, ch. 5). WVS allows us to examine the geographies of wellbeing over time, which is important as Poland has evolved greatly, and WVS data goes back to 1989. We have 4k observations, about 1k per wave.[14] Also, an important advantage of WVS is that urbanicity is recorded by an interviewer.[15]

Freedom is measured in WVS,[16] and in some ESS waves containing a whole battery of wellbeing items measuring functionings and vitality. Preliminary analyses using ESS 1-6 cumulative file already had over 10k observations for Poland.[17] We will use ESS items such as those in wave 6 on functioning "free to decide how to live my life," and vitality "had lot of energy."[18]

FB539 (2024) has 5.2k observations and is representative of 17 NUTS2 regions, and has a usual SWB question similar to WVS.[19] FB366,419 (2012, 2015) are representative of 4 Polish cities: Bialystok, Gdansk, Cracow, and Warsaw. The sample size is about 2000 observations per wave, and again we have a usual SWB question like in WVS.

Data harmonization is not needed in WVS–one advantage of the survey is consistent measurement over time. We will use the data file that has all the waves in it with consistent coding of the variables. FB539 is a cross section. FB366,419 are 2 repeated cross sections (2012, 2015)–we will check coding of the survey items and if they differ and if we make over time comparisons using these data, we will harmonize. The same pertains to ESS.

## 5.1 Models

We will mostly use ordinary least squares (OLS).[20] For robustness/sensitivity checks for the key models we will use multinomial logit (as the ordinal logit tends to fail proportional odds assumption). As another robustness/sensitivity check we are considering machine learning models such as gradient boosting models (Oparina et al. 2025), specifically XGBoost (`https://xgboost.readthedocs.io`).

We will use a usual set of controls such as those in the literature: for life satisfaction (e.g., Berry and Okulicz-Kozaryn 2009, Okulicz-Kozaryn 2010); and for freedom (e.g., Pitlik 2016, Nikolaev and Bennett 2016). A key robustness/sensitivity check will be specification curve analysis (SCA) (Simonsohn et al. 2020). And we will pay extra attention to the choice of control variables and avoid overcontrol bias.[21]

To examine geographic patterns among regions we will use choropleth maps, Moran's I, and Local Indicators of Spatial Association (LISA) (Anselin 1995).[22]

[14] We use waves 1989, 1997, 2005, and 2012. For the 1997, 2005, and 2012 waves we have 8 rural-urban categories -2k, 2-5k, 5-10k, 10-20k, 20-50k, 50-100k, 100-500k, and 500k-. In 1989 we only have information about 49 voivodeships, but urbanicity can be inferred from this information indirectly, and proximity to a major city directly.

[15] It is more objective than subjective respondent self-reports. For instance, respondents may not know well whether they live in a 20k or 30k place.

[16] "Some people feel they have completely free choice and control over their lives, while other people feel that what they do has no real effect on what happens to them. Please use this scale where 1 means 'none at all' and 10 means 'a great deal' to indicate how much freedom of choice and control you feel you have over the way your life turns out." Validity and reliability of the measure seem fine and the variable has been used successfully in the literature (e.g., Verme 2009, Inglehart et al. 2008, Veenhoven 2000a, Okulicz-Kozaryn 2014a).

[17] The data will be updated upto wave 12 that will be available in 2027.

[18] ESS urbanicity is measured with: a big city, suburbs or outskirts of big city, town or small city, country village, farm or home in countryside. Per wellbeing see overview at `https://www.europeansocialsurvey.org/findings/europeans-wellbeing/measuring-wellbeing`. Wave 6 wellbeing items: `https://europeansocialsurvey.org/sites/default/files/2026-01/ESS12-wellbeing-module-design-template.pdf` see Table 1: Items from the well-being module grouped by the dimension of wellbeing they relate to. We also will use new wave 12 that will be available in 2027 and will contain a wellbeing module. See `https://europeansocialsurvey.org/sites/default/files/2026-01/ESS12-wellbeing-module-design-template.pdf`.

[19] We may also use the earlier flash barometers: #497 (2021), #472 (2018), #427 (2015), #356 (2012) depending on how strong and robust the latest wave results are.

[20] Although OLS assumes cardinality of the outcome variable, and SWB measures are technically ordinal, OLS is an appropriate estimation method. Ferrer-i-Carbonell and Frijters (2004) has shown that OLS results are substantially the same as those from discrete models, and OLS has become the default method in happiness research (Blanchflower and Oswald 2011). Theoretically, while there is still debate about the cardinality of SWB, there are strong arguments to treat it as a cardinal variable (Ng 1996, 1997, 2011).

[21] "What we really need, to identify/estimate X− >Y, is to control for other determinants of Y that are also antecedents of X (so, W− >X). If we include controls where the relationship goes the other way (X− >W), we will exacerbate bias in our estimate of X− >Y. (The relevant term for that situation is 'overcontrol bias')" (Bartram et al. 2024, p. 5).

[22] We may do some multilevel modeling (MLM) at NUTS2 level as in Weckroth and Kemppainen (2021) but focusing on Poland. Yet MLM is problematic with only 17 U/As at Level 2–if we use it, results will be postponed to appendix. We may use Small Area Estimation (Moretti and Ahmed 2024).

# 6 Significance, Policy, and Practice

The proposed research will contribute to the literature and produce new knowledge. We will write at least 4 peer-reviewed articles[23] in this project, and through further research mentoring and leadership additional ones in the long term (see section "3.15. Didactic/mentoring activities" in the application form). Regarding the activities during the duration of this project see in this document section "Work Plan: Key Milestones."

The research also has policy and practical implications. First, as stated in the introduction, the time has come for Poles to focus on wellbeing as opposed to just survival. Poland has modernized infrastructure, higher incomes, and booming economy. The focus on survival and economy can now give a way to wellbeing and flourishing. With the proposed study we aim to aid in this pursuit of happiness. This is called evidence-based pursuit of happiness in the literature (Veenhoven 2012).[24] The proposed study will find out which parts of Poland enjoy high wellbeing and for whom–which social groups, e.g., by age and social class, flourish where? For instance, it may be the young and the educated that do not suffer from urban SWB penalty (Okulicz-Kozaryn 2014b). Our findings should help Poles increase their wellbeing.

There are many avenues for specific policy and practice collaborations. One simple implementation is to educate policy makers, public administrators, and citizenry about the wellbeing metrics and study's findings so that we can make more informed decisions. To that end we will share and popularize ourfindings through public lectures, popular press, and social media. In the long term we are thinking about establishing a wellbeing research center in Poland, similar to Erasmus Happiness Economics Research Organisation (EHERO; `https://www.eur.nl/ehero`), where PI used to work.

SWB can and should be used for public policy and administration (Diener 2009). SWB can serve as a "yardstick" (Veenhoven 1988)–arguably the ultimate outcome of public policy and administration is SWB.[25] "If you can't measure it, you can't improve it"–if governments don't measure wellbeing, they can't improve wellbeing. I will reach out to Polish government to implement the study findings in practice. I plan to start locally with Wielkopolska voivodeship, where I will be physically carrying out the study, and other regional governments may be involved as well. Likewise, some city governments may be involved as the study will use data representative of 4 cities.

It is relatively straightforward to implement the study's findings. For instance, we will find where and which social groups flourish and languish–such information can be then used to advance utilitarianism, produce the "greatest happiness for the greatest number" (Akay et al. 2015, Nussbaum 2012, Kahneman et al. 1997). Another avenue is to use the wellbeing metrics to evaluate policy. For instance, if there a major change in policy or investment in public goods, we can actually measure whether it produced an increase in wellbeing by using before-after comparative group research design (difference-in-difference). One avenue would be to engage with energy policy. While in general energy use has been found to be unrelated to SWB earlier (Okulicz-Kozaryn and Altman 2019), this claim needs to be re-examined amidst current energy crisis and skyrocketing prices. Massive reductions in consumption and altering of the lifestyles may ultimately reduce wellbeing. Finally, long term (after this project) it could be data collection, say implementing a survey similar to Flash Barometers locally; or using social media data to meeasure wellbeing (Iacus et al. 2025).

In framing policy and practice discussion we will also use Latin American perspective. One of the team members is a senior wellbeing scholar from Latin America, a public policy researcher whose work integrates the analysis of social programs, impact evaluation, and subjective well-being metrics. Our expert has also practical experience–has served as a consultant for the World Bank and the Inter-American Development Bank in Latin America. Latin America is the happiest region in the World (Rojas 2015, ????), and while there are similarities between Poland and Latin America, such as high levels of social connection, there are specific actionable lessons from

[23] Out of which at least 3 will be published in major international journals such as Social Indicators Research, Applied Research in Quality of Life, and Cities, with IF>3.

[24] There is a psychological nuance here–one should not focus fully on the pursuit of happiness and chase it or value it intensely (Mauss et al. 2011, Gruber et al. 2011), neither be concerned about it and judge it (Zerwas et al. 2024)–it may backfire. Rather, one needs to know what is important for wellbeing from the academic literature, like the proposed study, and include this information in everyday decision-making. For instance, humans should buy experience such as travel and time with others, not things such as conspicuous consumption items (Okulicz-Kozaryn and Tursi 2015, Veblen 2005a,b, Thoreau 1995 [1854]).

[25] While the idea of using wellbeing data for policy and administration is new in Poland, it is not new in general (Stiglitz et al. 2009, Diener 2009).

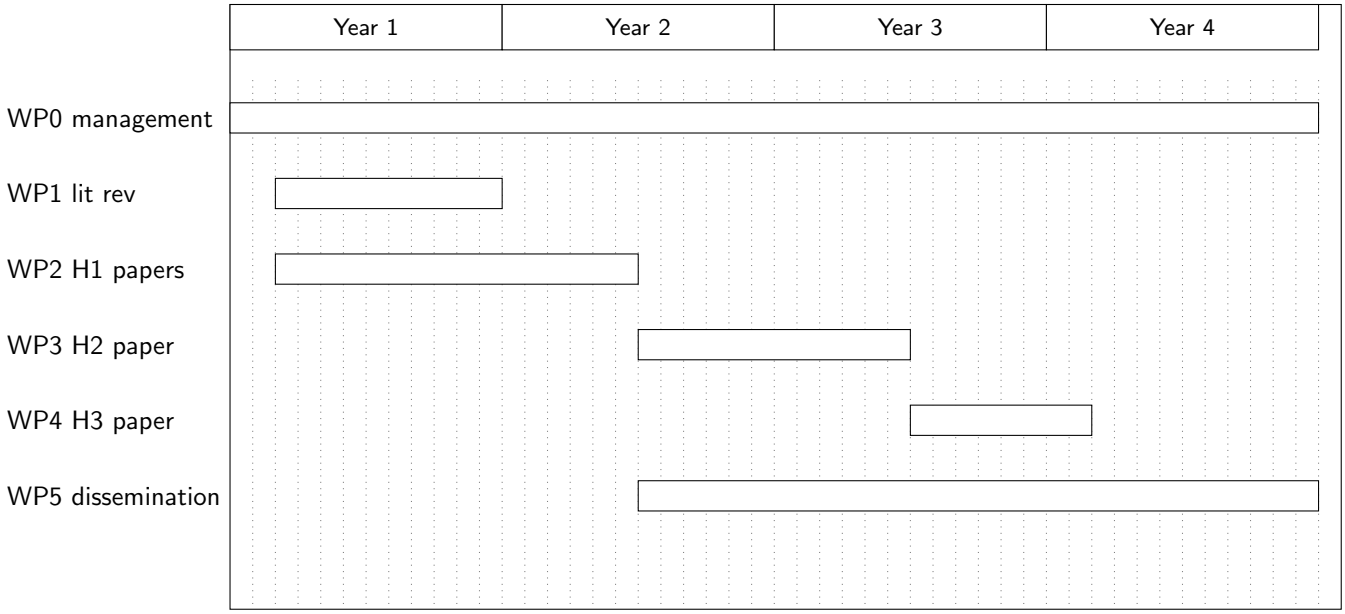
the region that other countries can use (Okulicz-Kozaryn and Martinez 2025).

# 7 Research Plan: Work Packages (WPs) and Tasks (Ts); and Work Plan: Key Milestones (KMs)

note: in square brackets are enumerated team members responsible for each task; order indicates workload from high to low

- all=all team members
- aok=adam okulicz kozartn (principal investigator)
- lm=lina marinez
- em=ebshoy mikhaeil
- pm=piotr matczak ???see katarzyna email and if ok TODO add to tasks
- mb=mariusz baranowski
- pj=piotr jablkowski
- pc=piotr cichocki
- student= AMU (Adam Mickiewicz University) graduate student; there are 3 of them
- WP0 monthts 0-48: management [aok]
  - T0 months 0-48 management: project management and coordination notably tracking the timeline and managing progress, communication, and risks [aok]
  - T1 months 0-2 meet and greet: conduct a series of in-person and zoom meetings to plan and get going with the project [all]
  - T2 months 0-2 setup project: complete necessary paperwork, notably employ team members at the university [all]; note: IRB approval is not necessary as we are using secondary data only
  - T3 months 0-48: in-person (whenever possible) or zoom (if necessary) weekly or at very least every two weeks meetings of team members currently involved in project tasks with PI [aok, team members currently working on tasks]
  - T4 months 45-48 report and finalize: write final report and finalize the project [aok,lm,em]
- KM0 month2: setup done
- WP1 monthts 2-12: literature review
  - T1: assign literature domains to 2 AMU students, and have students perform the review [2students]
  - T2: have more senior team members supervise the students and hold periodic meetings at least every few weeks or as necessary [em,aok]
  - draft and revise literature reviews for subsequent 4-5 journal articles [2students,em,aok]
- KM1 month12: literature review done
- WP2 months 2-18 H1 paper: data management, analysis, and paper drafting for hypothesis H1
  - T1 data management: clean, organize, document, recode [em,1student,aok]
  - T2 exploratory data analysis: descriptive statistics and visualizations [em,1student,aok]
  - T3 main analyses: mostly regressions and robustness checks (see for details methods section) [em,aok,1student]
  - T4 review and revisions of the analyses by the senior team members [aok,pc,mb,pj]
  - T5 drafting of the paper and revisions [aok,lm,em,2students]
- KM2 month18: data analyzed and papers drafted for H1
- WP3 months 18-30 H2 paper: data management, analysis, and paper drafting for hypothesis H2; Tasks are the same as in WP2
- KM3 month30: data analyzed and papers drafted for H2
- WP4 months 30-38 H3 paper: data management, analysis, and paper drafting for hypothesis H3; Tasks are the same as in WP2

- KM4 month38: data analyzed and papers drafted for H3
- WP5 months 18-48 dissemination: academic dissemination, real-world popularization, and policy engagement
  - T1 articles: $>=$ 3 submissions to journals with $IF > 3$ such as `Applied Research in Quality of Life`, `Social Indicators Research`, or `Cities` [all]
  - T2 conferences: 3 conference presentations such as `International Society for Quality-of-Life Studies (ISQOLS)` or `European Regional Science Association (ERSA)` [aok,lm,em]
  - T3 popularization via popular media (e.g., Glos Wielkopolski) and or social media (e.g., YouTube) [aok,lm,em,1student]
  - $>=$4 public lectures and or seminars at AMU and or other Polish universities
  - T4 engage with local public policy and administration starting in Poznan and Wielkopolska [aok,1student]
- KM5 month48: dissemination and popularization done




## 8 Risks

see application form "3.9. Risk analysis and security issues"

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